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RAPID REPORT

Sex differences in mitochondrial respiratory function in human skeletal muscle

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Miotto PM, McGlory C, Holloway TM, Phillips SM, Holloway GP. Sex differences in mitochondrial respiratory function in human skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 314: R909–R915, 2018. First published March 7, 2018; doi:10.1152/ajpregu.00025.2018.—Mitochondrial bioenergetic contributions to sex differences in human skeletal muscle metabolism remain poorly defined. The primary aim of this study was to determine whether mitochondrial respiratory kinetics differed between healthy young men and women in permeabilized skeletal muscle fibers. While men and women displayed similar ($P > 0.05$) maximal respiration rates and abundance of mitochondrial/adenosine diphosphate (ADP) transport proteins, women had lower ($P < 0.05$) mitochondrial ADP sensitivity (+30% apparent K_m) and absolute respiration rates at a physiologically relevant ADP concentration (100 μM). Moreover, although men and women exhibited similar carnitine palmitoyl transferase-I protein content- and palmitoyl-CoA-supported respiration, women displayed greater sensitivity to malonyl-CoA-mediated respiratory inhibition. These data establish baseline sex differences in mitochondrial bioenergetics and provide the foundation for studying mitochondrial function within the context of metabolic perturbations and diseases that affect men and women differently.

ADP sensitivity; aerobic metabolism; lipid metabolism; mitochondria; sex differences; skeletal muscle

INTRODUCTION

Humans display sex differences in skeletal muscle metabolism that may be due to genetic (34), environmental, and/or hormonal influences (37). While an explanation for these differences remains poorly defined, elevated 17-β estradiol in women is believed to promote the induction of genes regulating lipid metabolism and mitochondrial function. In support of this, women have a greater proportion of type I muscle fibers (4, 8, 9, 35, 43), higher intramuscular lipid stores (33), and greater mRNA expression of genes associated with fatty acid metabolism, including cluster of differentiation-36 (CD36)/fatty acid binding protein (FABPpm) (19), hormone-sensitive lipase (33), carnitine palmitoyl transferase-I (CPT-I), β-hydroxyacyl-CoA dehydrogenase (β-HAD) (23), and citrate synthase (23), while short-term supplementation of 17-β estradiol in men increases peroxisome proliferator-activated receptor γ coactivator 1-α (PGC-1α) and the expression of proteins involved in mitochondrial β-oxidation (22). Moreover, lower 17-β estradiol in aging women is associated with a greater risk of developing chronic diseases linked with impaired mitochondrial metabolism, including heart disease and Type 2 diabetes (11). Altogether, these data implicate alterations in mitochondrial bioenergetics as a contributor to metabolic sex differences.

Although 17-β estradiol has the potential to influence mitochondrial gene transcription, men and women do not display differences in mitochondrial enzymatic function, density, or morphology (38), suggesting an alternative explanation. Instead, this hormone may influence the intrinsic function of the mitochondria. In this regard, 17-β estradiol has recently been shown in rodents to positively regulate mitochondrial bioenergetics, independent of mitochondrial content or maximal enzymatic rates, by directly influencing mitochondrial membrane viscosity (39). Therefore, sex differences in mitochondrial bioenergetics in humans may exist despite similar mitochondrial protein expression and enzymatic capacity.

A prominent control point for regulating aerobic metabolism involves the transport of ADP and lipids through the mitochondrial voltage-dependent anion channel (VDAC)/adenine-nucleotide translocase (ANT) axis (12) and CPT-I (24), respectively. Given that membrane fluidity is believed to influence the structure/function of integral membrane proteins, and 17-β estradiol has been linked to mitochondrial membrane viscosity (39), we hypothesized that women would exhibit respiratory substrate kinetics conducive toward greater mitochondrial substrate transport/metabolism. To examine this, we determined muscle mitochondrial and substrate transporter protein expression, mitochondrial respiration, and substrate sensitivity between healthy young men and women.

METHODS

Participants and skeletal muscle biopsy. Sixteen healthy moderately trained men ($n = 8$) and women ($n = 8$) that reported to be recreationally active 2–3 times per week were recruited for the study. Individuals arrived at the laboratory in the fasted state, body mass and height were recorded, and a dual-energy X-ray absorptiometry scan was performed. Thereafter, participants were rested in a semi-supine position on a bed, the skin surrounding the vastus lateralis muscle was cleaned, local anesthetic was applied (2% lidocaine, 0.005 mg/ml epinephrine), and an incision was made to the skin and fascia. A sample of vastus lateralis (~30 mg) was extracted using a Bergström needle and immediately placed in ice-cold BIOPS for preparation of permeabilized muscle fibers (described below). All participants provided informed written and oral consent before commencing the study, and all procedures were approved by the Hamilton Integrated

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Research Ethics Board at McMaster University, and conformed to the most recent version of the Declaration of Helsinki.

Preparation of permeabilized muscle fibers. As previously validated (40), permeabilized muscle fibers were separated under a microscope in preservation buffer (BIOPS, in mM: 50 MES, 7.23 K_2EGTA, 2.77 CaK_2EGTA, 20 imidazole, 0.5 DTT, 20 taurine, 5.77 ATP, 15 PCr, and 6.56 $MgCl_2 \cdot H_2O$; pH 7.1). Fibers were treated with saponin (30 μg/ml) (18) to permeabilize the sarcolemma and were then washed in MIRO5 (0.5 mM EGTA, 3 mM $MgCl_2 \cdot 6H_2O$, 60 mM potassium lactobionate, 10 mM KH_2PO_4, 20 mM HEPES, 110 mM sucrose, and 1 g/l fatty acid-free BSA; pH 7.1) until respiration analyses.

Mitochondrial respiration in permeabilized muscle fibers. Mitochondrial respiration measurements were performed in MIRO5 using high-resolution respirometry (Oroboros Oxygraph-2K, Innsbruck, Austria) at 37°C. All experiments were conducted between 200 and 180 μM of oxygen, and the buffer was equilibrated to room air after the addition of each substrate. All experiments were performed in the presence of 5 μM blebbistatin, a myosin II inhibitor. ADP-stimulated respiration was determined by titrating ADP (0, 100, 250, 500, 1,000, 2,000, 4,000, 6,000, 8,000, 10,000, 12,000, 16,000, 18,000 μM) until a plateau was attained in the presence of saturating pyruvate (5 mM) and malate (5 mM). Subsequently, glutamate (10 mM) and succinate (10 mM) were added at the end of experiments to determine maximal complex I- and II-supported respiration, respectively. In separate fiber experiments, palmitoyl-CoA (P-CoA)-stimulated respiration (25 and 50 μM) was determined in the presence of malate (5 mM), ADP (20 mM), and L-carnitine (2 mM). To determine percent inhibition as a surrogate of CPT-I sensitivity to malonyl-CoA (M-CoA), M-CoA (7 μM) was added on the back end of experiments. Respiratory control ratios (RCRs) were calculated by dividing state 3 (+ADP)/state 2 (−ADP) respiration. Cytochrome *c* (10 μM) was added at the end of all experiments to ensure mitochondrial integrity and did not stimulate mitochondrial respiration (data not shown). All fiber bundles were recovered and freeze dried, and data were normalized to fiber weight.

Fiber digestion and Western blot analysis. Western blot analysis was performed on remaining nonpermeabilized skeletal muscle from the biopsy. Muscle was freeze-dried, weighed, and digested (5 μl/μg) in a buffer containing glycerol (10%), β-mercaptoethanol (5%), SDS (2.3%), and Tris·HCl (62.5 mM; pH 6.8) (13). Samples were then loaded equally for OXPHOS (1:500, cat. no. ab110413; Abcam), pyruvate-dehydrogenase-subunit E1α (PDHE1α, 1:5,000, cat no. 459400; Invitrogen), pyruvate dehydrogenase kinase-4 (PDK4, 1:1,000, cat. no. ab38242; Abcam), PGC-1α (1:1,000, cat. no. 516557; Calbiochem), peroxisome proliferator-activated receptor γ coactivator-β (PGC-1β, 1:1,000, cat. no. NB110–58858; Novus), sirtuin-1 (SIRT1, 1:1,000, cat. no. 2493; Cell Signaling), histone acetyltransferase-5 (GCN5, 1:1,000, cat. no. 607202; Biolegend), VDAC (1:5,000, cat. no. ab14734; Abcam), mitochondrial creatine-kinase (Mi-CK, 1:5,000, cat. no. ab131188; Abcam), ANT1 (1:1,000, cat. no. ab110322; Abcam), FABPpm, (1:30,000), CD36 (1:1,000, sc13572; Santa Cruz), adipose-triglyceride-lipase (ATGL, 1:1,000, cat. no. 10006409; Cayman), Lipin 1 (1:1,000, cat. no. ab84950; Abcam), glycerol-3-phosphate acyltransferase (GPAT; 1:1,000; Santa Cruz), diglyceride acyltransferase (DGAT; 1:1,000, cat. no. IMG-30279; Imgenex), CPT-I (1:1,000, cat. no. CPT1M11-A; Alpha-diagnostics), and β-HAD (1:1,000, cat. no. ab154088; Abcam). CIV is absent from OXPHOS images as a result of the digestion protocol, as previously reported (13). Antibody validation for these targets have been previously reported (1, 3, 5, 6, 14, 25–29, 31, 32, 44). Proteins were separated using SDS-PAGE, blocked, and incubated with the appropriate primary and secondary antibodies. All membranes were detected using enhanced chemiluminescence (ChemiGenius2 Bioimaging system; SynGene).

Statistical analyses. Michaelis-Menten kinetics were determined using GraphPad Prism 7 software. Sex differences were determined using Student's independent samples *t*-test (SigmaPlot 11.0, Systat Software) between men and women. The *P* values were two-tailed for all parameters with the exception of CD36 that has been previously reported to exhibit sex differences (one-tailed). Associations between fat mass and mitochondrial bioenergetics were performed using Pearson *r* correlational analyses. Significance was determined as $P < 0.05$. Data are expressed as means ± SE.

RESULTS

Participant characteristics. Men and women were matched for age and body mass index, and therefore, there were no significant differences in these parameters (Table 1). As expected, women had a lower height, lower body mass, and lean mass, and greater fat mass than men ($P = 0.09$ for absolute fat mass; $P = 0.002$ relative fat mass (Table 1).

Women have lower ADP sensitivity than men independent of maximal respiration rates and mitochondrial protein expression in skeletal muscle. Given that 17-β estradiol has recently been shown to regulate mitochondrial respiration in rodents (39), we first assessed maximal respiration rates in men and women. RCR values above 4 confirm viable fiber preparations in both men and women (Fig. 1*A*). Men and women had similar mitochondrial respiration rates in the presence of complex I (pyruvate, glutamate)- and II (succinate)-linked substrates, as well as maximal ADP-stimulated respiration (Fig. 1*A*). In addition, men and women displayed similar abundance of mitochondrial proteins, including various electron transport chain complexes (Fig. 1*B*), and similar levels of key proteins involved in transcriptional regulation of mitochondrial biogenesis, including PGC-1α, PGC-1β, SIRT1, and GCN5 (Fig. 1*B*).

We next examined mitochondrial ADP sensitivity, as we reasoned this could be regulated independent of mitochondrial content. While men had comparable apparent K_m values as the existing literature for ADP (13, 21, 30), women had significantly lower ADP sensitivity (~30%; greater apparent K_m) (Fig. 2, *A* and *B*) and lower absolute respiration at a physiological (100 μM) concentration of ADP (men: 82.53 ± 17.69; women: 47.50 ± 2.56, $pmol \cdot s^{-1} \cdot mg\ dry\ wt^{-1}$, $P < 0.05$) than men. This occurred independent of alterations in the content of ADP transport proteins VDAC, Mi-CK, and ANT1 (Fig. 2*C*). Altogether, these data highlight sex differences in the regulation of the VDAC/ANT axis, independent of mitochondrial capacity measurements.

Women have similar P-CoA-stimulated respiration rates and greater sensitivity to M-CoA than men in skeletal muscle. We next examined indexes of mitochondrial fatty acid oxidation (e.g., CPT-I-dependent respiration) and protein content of markers involved in lipid handling between men and women. There was no difference in absolute P-CoA-stimulated respiration rates in the presence of P-CoA (25 or 50 μM) between

Table 1. *Participant characteristics*

	Men	Women	*P* Value
Age, yr	23 ± 1	21 ± 1	0.193
Height, m	1.8 ± 0.02	1.6 ± 0.02	<0.001
Total mass, kg	76.1 ± 2.0	63.2 ± 3.5	0.006
BMI	23.8 ± 0.62	24.1 ± 1.6	0.865
Body fat, %	19 ± 2	32 ± 3	0.002
Body fat, kg	14.0 ± 1.4	20.0 ± 3.1	0.09
Lean mass, kg	59.2 ± 1.9	40.5 ± 1.5	<0.001

Data are expressed as means ± SE. BMI, body mass index.

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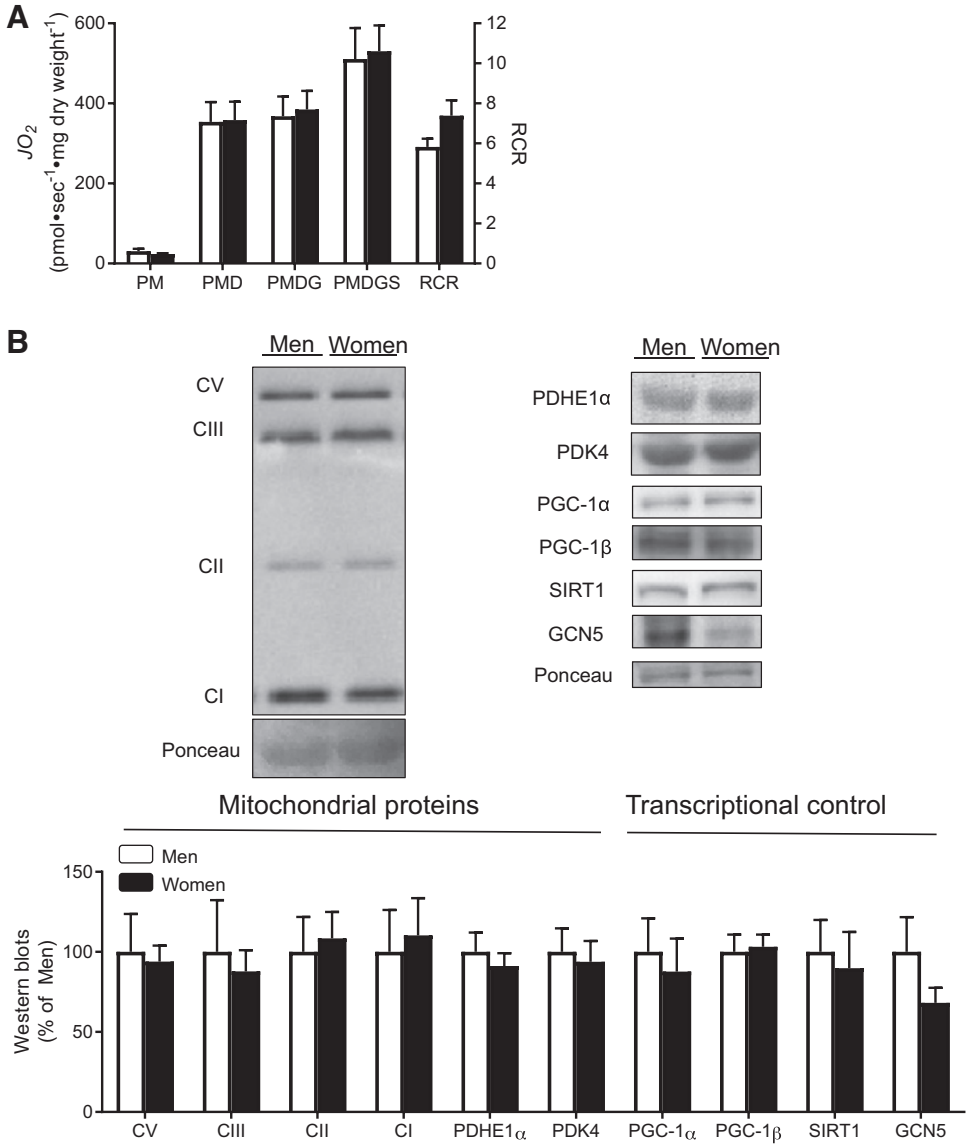


Fig. 1. Maximal mitochondrial respiration and markers of mitochondrial content. Rates of oxygen consumption (JO_2; *A*) in permeabilized muscle fibers were determined in men and women with the sequential addition of pyruvate+malate (PM, Complex I), ADP (PMD), glutamate (PMDG, maximal Complex I), and succinate (PMDGS, Complex I+II). Respiratory control ratio (RCR) is complex I respiration in the presence and absence of ADP. *B*: Western blots were used to estimate the abundance of mitochondrial proteins, as well as proteins involved in regulating mitochondrial biogenesis. GCN5, histone acetyltransferase-5; PDHE1α, pyruvate-dehydrogenase-subunit E1α; PDK4, pyruvate dehydrogenase kinase-4; PGC-1α, peroxisome proliferator-activated receptor γ-coactivator 1-α; PGC-1β, peroxisome proliferator-activated receptor-γ coactivator 1-β; SIRT1, sirtuin-1. C denotes the complex of various subunits of the electron transport chain (CI, CII, CIII, and CV). *Significant difference between men and women ($P < 0.05$). Data are expressed as means ± SE; $n = 7$ or 8 participants for each sex.

men and women (Fig. 3*A*). Given the observation that women had slightly lower respiration rates compared with men before the addition of P-CoA, we also expressed the data as percent stimulation to examine the direct ability of P-CoA to stimulate respiration. When we used this approach, there were still no differences in percent stimulation on P-CoA respiration between men and women (Fig. 3*B*), further suggesting a similar capacity for lipid-supported respiration. In contrast, the ability for M-CoA to inhibit P-CoA-stimulated respiration was greater in women (Fig. 3*B*), implicating a greater sensitivity of CPT-I to M-CoA inhibition in human skeletal muscle of women. While this response would predict lower CPT-I flux in women, in vivo, fatty acid oxidation is also influenced by substrate availability. Therefore, we next examined markers of lipid transport, synthesis/lipolysis, and oxidation. Supporting the P-CoA respiration data, CPT-I and β-HAD protein content were similar between sexes (Fig. 3*C*). While there were no differences in proteins involved in lipid synthesis/lipolysis (ATGL, Lipin 1, GPAT, DGAT) or FABPpm, a key plasma membrane lipid transporter (CD36) was higher in women (Fig. 3*C*), which could contribute to sex differences in intramuscular lipid metabolism by increasing substrate availability.

Association between body fat mass and mitochondrial bioenergetics. Since the women in the current study had greater fat mass than men, and previous work has highlighted a relationship between fat mass and muscle mitochondrial capacity (16), we next used Pearson *r* correlational analyses to determine the relationship between mitochondrial kinetics and fat mass. There was a trending positive association between percent fat mass and apparent K_m for ADP ($r = 0.477$; $P = 0.06$). Similarly, there was a trend for a positive relationship between percent fat mass and M-CoA inhibition ($r = 0.348$; $P = 0.1$).

DISCUSSION

Our data provide novel insight that women have moderately lower mitochondrial ADP sensitivity and greater M-CoA sensitivity than men. This occurred independent of maximal respiration and abundance of mitochondrial proteins or substrate

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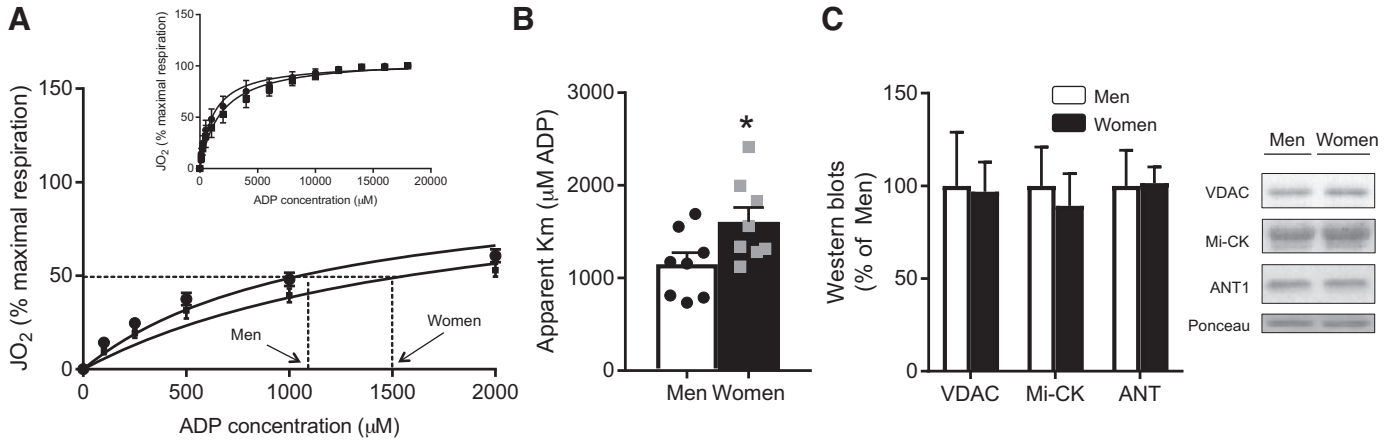


Fig. 2. Mitochondrial ADP respiratory sensitivity. Rates of oxygen consumption (JO_2) in permeabilized muscle fibers were determined in men and women in the presence of pyruvate+malate and various concentrations of ADP. These values followed a Michaelis-Menten kinetic curve (*A*; *inset* shows the full kinetic curve; circles denote men and squares denote women), which was used to estimate ADP sensitivity (apparent ADP K_m; *B*). Western blots were utilized to estimate the abundance of mitochondrial proteins involved in ADP transport (*C*). ANT1, adenine nucleotide translocase-1; Mi-CK, mitochondrial creatine-kinase; VDAC, voltage-dependent anion channel. *Significant difference between men and women ($P < 0.05$). Data are expressed as means ± SE; $n = 8$ (respiration analyses) and $n = 7$ or 8 (Western blots) participants for each sex.

transporters in resting skeletal muscle. Altogether, these data do not support the supposition that women display higher mitochondrial membrane substrate transport.

Sex differences in skeletal muscle ADP sensitivity. Although men and women are known to have different skeletal muscle metabolic properties (4, 8, 9, 35, 43), whether this divergence extends to mitochondrial function and substrate transport capacity remains poorly defined. However, this may be an important consideration for understanding general skeletal muscle metabolism, given that the transport of ADP through the VDAC/ANT axis could modulate cytosolic ADP concentrations and glycolytic flux (42), and it is already well established that men and women have different fuel reliance depending on energy demand (4, 10, 15). In our current study, the similar maximal respiration rates suggest any potential hormonal differences between men and women do not alter mitochondrial capacity, which is further supported by the similar protein expression of electron transport chain complexes. While previous work in mice has shown that ablation of circulating 17-β estradiol was detrimental for maximal respiratory flux (39), our current data suggest that women, who typically have two- to threefold higher 17-β estradiol levels (17), do not display different respiratory capacity than men.

Another major aim of the present study was to examine potential sex differences in mitochondrial ADP sensitivity. Although we hypothesized that women would have greater respiratory kinetics in response to ADP, women, in fact, displayed lower ADP sensitivity. The lower mitochondrial ADP sensitivity and respiration rate at 100 μM ADP in women occurred in the presence of similar levels of VDAC, Mi-CK, and ANT1, suggesting a possible sex-specific regulation of ADP sensitivity. Although the functional role for the lower ADP sensitivity remains to be determined and may be influenced by a potential relationship with fat mass, this lower ADP sensitivity does not likely contribute to the improvement in aerobic metabolism or fatty acid oxidation observed in women compared with men (7).

Sex differences in CPT-I regulation by M-CoA in skeletal muscle. It is well established that women, or men supplementing with 17-β estradiol, rely more heavily on lipid oxidation during exercise (37). Although the mechanisms of action for these sex differences in lipid metabolism remain poorly defined, an increase in lipid delivery to muscle has been proposed as a contributing factor. In addition to women displaying higher lipoprotein lipase (19) and intramuscular lipid content (33), our data demonstrate that women have higher CD36 protein expression, as observed previously (19). Surprisingly, despite fatty acid metabolism being a mitochondrial localized process, direct comparisons between mitochondrial respiratory function between sexes have been lacking. In the present study, we show that men and women display similar CPT-I content and P-CoA-stimulated respiration rates. Importantly, we used submaximal concentrations of P-CoA that reflect physiological muscle concentrations (41) and revealed similar responses among men and women, supporting previous findings of unaltered maximal CPT-I activity (2). In addition, we present a novel finding that women have greater sensitivity for M-CoA inhibition of P-CoA-stimulated respiration. Although speculative, the greater M-CoA sensitivity in women may be required to sustain low-energy demands at rest and turn down mitochondrial substrate flux, especially given that women in the current study had greater percent fat mass, a trending relationship between percent fat mass and M-CoA sensitivity, and, typically have greater intramuscular lipid stores (33). Regardless, the present finding of greater M-CoA sensitivity and similar P-CoA sensitivity suggest greater CPT-I flux may not contribute to sex differences in humans.

Perspectives and Significance

Overall, we provide novel insight that men and women have similar maximal respiration rates and different substrate sensitivity independent of protein expression of substrate transporters, lipid handling, or mitochondrial capacity in resting skeletal muscle. Within this context, an important consideration is participant training status, as exercise training can

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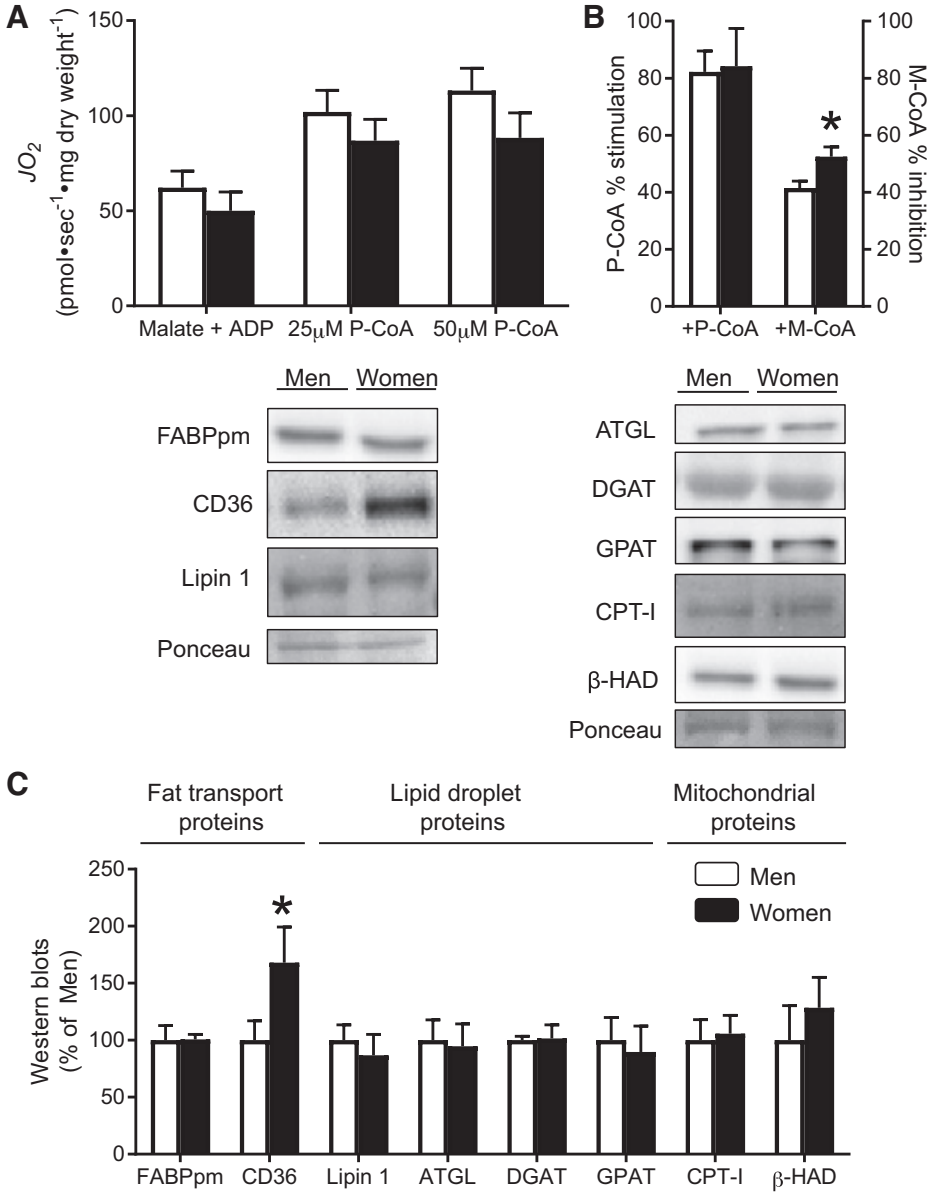


Fig. 3. Mitochondrial lipid supported respiration and proteins involved in fatty acid metabolism. *A*: rates of oxygen consumption (JO_2) in permeabilized muscle fibers were determined in men and women in the presence of 25 and 50 μM palmitoyl-CoA (P-CoA). *B*: ability of P-CoA to stimulate respiration and malonyl-CoA (M-CoA) to inhibit respiration was also determined. *C*: Western blots were used to estimate the abundance of proteins involved in fatty acid metabolism. ATGL, adipose-triglyceride-lipase; CD36, cluster of differentiation-36; CPT-I, carnitine palmitoyl transferase-I; DGAT, diacylglycerol acyltransferase; FABPpm, fatty acid binding protein; GPAT, glycerol-3-phosphate acyltransferase; β-HAD, β-hydroxyacyl-CoA dehydrogenase. *Significant difference between men and women ($P < 0.05$). Data represent means ± SE; $n = 5–8$ participants for each sex in *A* and *B* due to an inability to recover some fibers to obtain accurate weights after respiration experiments were performed; $n = 7$ or 8 participants for each sex in *C*.

decrease ADP sensitivity (21). While women displayed lower ADP sensitivity indicative of greater physical fitness, training can also decrease M-CoA sensitivity (36), which is the opposite response that would be expected due to the observed sex differences. Therefore, it is unlikely that training status can fully explain the present data, especially considering both sexes had similar maximal respiration and mitochondrial content, parameters that typically increase together with training and positively correlate with maximal mitochondrial oxygen consumption in humans (20). Another potential confounding variable relates to skeletal muscle fiber type, as women are known to have a greater abundance of type I muscle fibers (4, 8, 9, 35, 43), and in rodents, oxidative muscles display lower mitochondrial ADP sensitivity (45). It remains unknown what accounts for this fiber type difference, and future work should determine whether type I fibers display different isoforms of ADP transport proteins (i.e., ANT/VDAC). Nevertheless, the present data provide compelling evidence that women display lower mitochondrial ADP sensitivity and greater M-CoA sensitivity, and given the relationship between mitochondrial content and function and various chronic diseases (11), these baseline sex differences provide the foundation for studying the role of mitochondrial bioenergetics within the context of metabolic perturbations and diseased states.

GRANTS

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

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AUTHOR CONTRIBUTIONS

P.M.M., S.M.P., and G.P.H. conceived and designed research; P.M.M. performed experiments; P.M.M. analyzed data; P.M.M., C.M., T.M.H., S.M.P., and G.P.H. interpreted results of experiments; P.M.M. and G.P.H. prepared figures; P.M.M. and G.P.H. drafted manuscript; P.M.M., C.M., T.M.H., S.M.P., and G.P.H. edited and revised manuscript; P.M.M., C.M., T.M.H., S.M.P., and G.P.H. approved final version of manuscript.

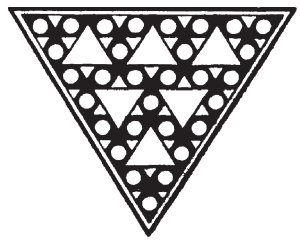
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